CHAPTER 12

EXPLORING SPACE

Prior to the recent scientific developments in the field of space science and technology, ground-based telescopes, spectroscopes and other similar devices were used to observe the sky and other objects in it. Sputnik-1 was the first man-made spacecraft, which was launched by the Soviet Union on October 4, 1957. With this, began the era of human space exploration wherein the human beings not only travelled into space, landed on the Moon but also returned safely. Today, the space exploration provides us numerous benefits such as better understanding of the Universe, socio-economic developments, technological growth and other advantages associated with it. In this chapter, we will study how space exploration has changed our daily lives.

In this chapter you will learn about:

- Telescope, Spectroscope, Spacecrafts.
- Space Exploration.

All the students will be able to:

- ✓ Describe development of tools and technologies used in space exploration.
- ✓ Analyze the benefits generated by the technology of space exploration.
- ✓ Explain how do astronauts survive and research in space.
- ✓ Suggest ways to solve the problems that have resulted from space exploration.
- ✓ Identify the technological tools used in space exploration.
- ✓ Identify new technologies used on earth that have developed as a result of the development of space technology.
- ✓ Design a space craft and explain its key features of design to show its suitability as a space craft.

Telescope, Spectroscope and Spacecrafts

- ✓ Describe the development of tools and technologies used in space exploration.

Telescope:

The word "*telescope*" is a combination of two Greek words: "Tele", which means 'distant or away' and "scope", which means 'to see'. Therefore, **telescope** can be defined as an instrument that enables us to see distant objects. Galileo Galilei- an Italian astronomer in the seventeenth century observed Jupiter and its four moons (i.e. Io, Europa, Ganymede and Callisto), Saturn and Venus with the help of a telescope first time in the human history.

Types of Telescope:

There are two types of a telescope: (i) Refracting Telescope, (ii) Reflecting Telescope.

Refracting Telescope:

A telescope that uses lenses is called a **Refracting Telescope**. There are two lenses in a Refracting Telescope: one is called Primary or Objective Lens, whose diameter is large while the other is called Secondary or Eyepiece Lens whose diameter is small. It consists of two tubes that slide into each other. Both the lenses are placed at the outer edges of the tubes. The primary lens focuses on the incoming rays of light that create an image. We see this image with the help of Secondary or Eyepiece Lens. Figure 12.1 shows the ray diagram of a Refracting Telescope.

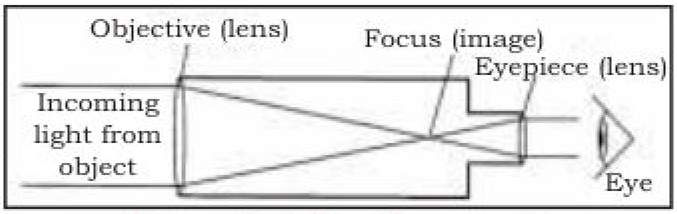


Figure 12.1: Ray diagram of a Refracting Telescope

Reflecting Telescope:

A telescope that uses mirrors is called a Reflecting Telescope. There are two mirrors in a Reflecting Telescope: one is called Primary or Objective Mirror whose diameter is large and the other is called Secondary Mirror whose diameter is small. A Reflecting Telescope comprises a single tube in which the Objective Mirror is placed at the rear end of the tube. It reflects the rays of light on the secondary mirror, which re-directs them towards eyepiece where an image can be seen. Figure 12.2 shows the ray diagram of a Reflecting Telescope.

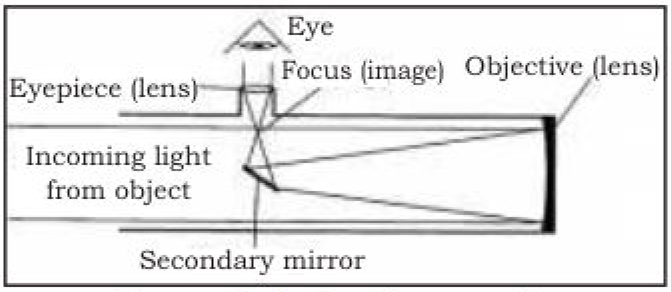


Figure 12.2: Ray diagram of a Reflecting Telescope

Telescopes have not only helped us in better understanding the astronomical objects in the sky but also expanded the horizon of Universe for further exploration and research. They also facilitate us in discovering new objects in the space every now and then. Today, a large number of telescopes have been installed in different countries of the world; while, at the same time, many other have also been sent into space. Currently, the Hubble Space Telescope (HST) is one of the most famous telescope launched in space. It is a Reflecting Telescope that has been sent into space as a joint venture of the US and European countries. Orbiting at an altitude of 600 kilometers from Earth, this telescope has provided us many invaluable images of different galaxies, clusters of stars, nebulae, etc.

Spectroscope:

A **spectroscope** is an optical instrument, which is used to measure the properties of visible light. It splits white light into its seven different component colours such as Violet, Indigo, Blue, Green, Yellow, Orange, and Red that are arranged according to their wavelengths in the spectrum of light. The set of colours obtained in this way is called a spectrum of light. From Figure 12.3, we can see that Red colour has the largest wavelength whereas Violet has the smallest.

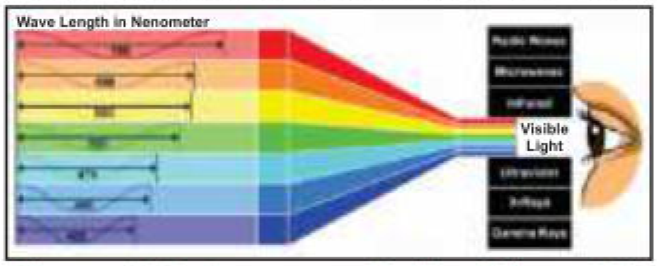


Figure 12.3: Spectrum of white light and its wavelength distribution

The construction and working of a Spectroscope (see Figure 12.4) is shown as follows:

- **Opaque Barrier with a Slit**: It forms a narrow beam of light.
- **Prism**: It splits the narrow beam of light into its seven component colours.
- **Detector or a Screen**: It allows the user to view the resulting spectrum of light.

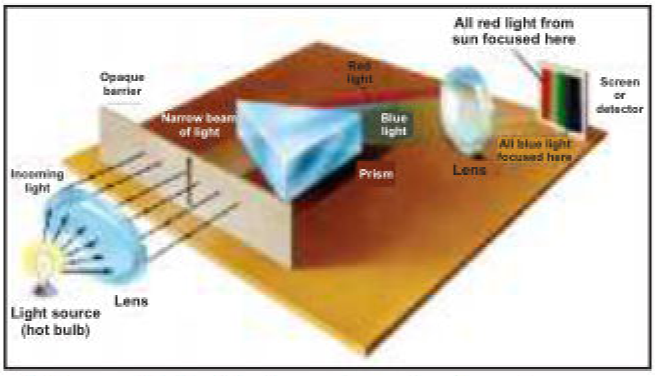


Figure 12.4: Main parts of a Spectroscope and their working

A Spectroscope is attached with a telescope to create a spectrum of light coming from a star. They are used in identifying Chemical Elements in a stellar atmosphere such as carbon, nitrogen, oxygen, etc. Thus, a spectroscope can tell us what elements are present in a star.

Spacecraft:

We use spacecraft for global navigation and communication, monitoring of weather, exploration of planets and other heavenly bodies. A **spacecraft** is a vehicle sent into space to carry out a specific task. A spacecraft can be a manned mission to transport humans and cargo into space and back to earth. It can also be an artificial satellite or a space probe commonly called unmanned spacecraft that is sent into space to gather precise data. Examples of manned spacecrafts are space shuttles namely Soyuz, International Space Station (ISS) and Apollo -17 Command Module that took humans to the Moon (see Figure 12.5).

Soyuz Space Craft

Apollo-17 Command Module

International Space Station (ISS)

Figure 12.5 Examples of manned spacecrafts

The Hubble Space Telescope, Venera - 9 and Opportunity Rover are some of the examples of un-manned spacecrafts.

Vanera-9

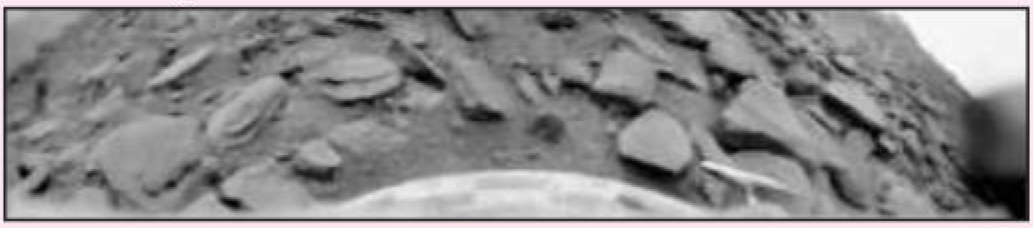
Opportunity Rover on the surface of Mars

Hubble Space Telescope (HST)

Figure 12.6: Examples of un-manned spacecrafts

DO YOU KNOW?

Venera-9 was the first spacecraft landed on surface of Venus. It was launched by Soviet Union in 1975. It became the first spacecraft to go into orbit around Venus.



Space Exploration:

- ✓ Analyze the benefits generated by the technology of space exploration.
- ✓ Identify new technologies used on earth that have developed as a result of the development of space technology.

Use of astronomy and space technology to explore space is called space exploration. Humans have always been curious about the night sky and outer space. With the passage of time, the curiosity of human nature and technological developments have enabled human beings to explore it physically by using manned and un-manned spacecrafts. It is expected that soon space exploration will pave a way for colonizing space to ensure the survival of the human race beyond this planet.

Benefits of Space Exploration:

More than fifty years of space exploration has provided numerous benefits, which have left a far-reaching impact on the daily life of the people living on Earth. The benefits of space exploration can be categorized as either direct or indirect benefits. The direct benefits of exploration include the generation of scientific knowledge, the diffusion of innovation and creation of commercialization opportunities, etc. Indirect benefits include tangible improvement in the quality of life such as economic prosperity, health, safety and security. A few of them are discussed as follows:

1. Health and Medicines:

- **Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) or Computerized Axial Tomography (CAT) Scans:** These are digital image processing procedures that are used to take images of inner limbs/parts of human body e.g. human brain. These methods were previously designed to enhance the pictures of the surface of the Moon.
- **Left Ventricular Assist Device (LVAD):** It is an artificial heart pump designed basing on the space shuttle's fuel pumps. It is used as an intermediatory measure to keep the heart patients alive before their heart transplant operations in the hospitals.
- **Breast Biopsy System**: It is an image-guide needle developed focusing on the technology used in the Hubble Space Telescope. It is used to obtain a sample from the abnormal development in the human breast for further laboratory tests.

- **Polyurethane Foam**: It is foam like material used to protect and insulate the external fuel tanks of the space shuttle. This foam is used to prepare less expensive molds for its use in the preparation and designing of artificial arms and legs for disabled and handicapped people.
- **Cooling Suits:** Liquid Cooling and Ventilation Garment technology is used in space suit to maintain a comfortable core body temperature of the astronautics during Extra Vehicular Activity (EVA). Basing on this technology, Cooling Suits are designed as a wearable garment to protect a patient's brain and other vital organs following a cardiac arrest.
- **Voice-Controlled Wheelchairs**: It's a voice-controlled wheel chair that is used for physically disabled people who cannot control the movements of their hands. It's designed basing on the tele-operator and robot technology used in the space programmes.
- **Light-emitting Diodes (LED)**: It's a special lighting technology, which has been developed for space based commercial plants growth in NASA's space shuttle. This technology is used to treat patients suffering from brain cancer.
- **Cataract Surgery Tool:** It's a tiny cutter pump designed by NASA as a part of its space technology. It is used to treat the eye patients suffering from the cataract disease.

2. Global Positioning System:

Global Positioning System (GPS) is a scientific method, which monitors the movement of vehicles, ships and aircrafts. It determines their location, route and distance travelled from one place to the other. It also provides real-time position information of the moving or static objects in all weathers across the globe.

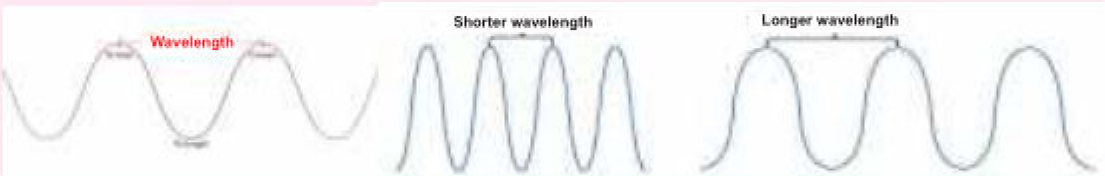
Figure 12.7: Artistic view of GPS satellites in orbiting around the Earth

The GPS consists of 30 or more satellites orbiting the Earth in the Medium Earth Orbit (MEO) that ranges from a few hundred miles to a few thousand miles above the surface of Earth. A GPS receiver on Earth receives signals from satellites and calculates its absolute position on Earth. Each satellite makes two complete orbits in 24 hours in such a way that at any time and anywhere on Earth, at least four satellites are always visible in the sky.

DO YOU KNOW?

What is wavelength?

Wavelength is defined as the distance between two consecutive upper peaks also called crests or lower peaks also called troughs of a wave. If crests or troughs are closer then the wavelength will be smaller or vice versa.



3. Weather Prediction/Forecast:

Weather forecasting means predicting or guessing about weather likely to happen in the near future by using different Weather Satellites. Scientists commonly known as Meteorologists do weather forecasting by continuously tracking and predicting the path of tornadoes, hurricanes, or floods. They take pictures of the Earth from space to carefully monitor weather conditions at any location around the world. Meteorologists warn us to seek shelter from dangerous and extreme weather conditions or hazards.

Figure 12.8: A Weather Satellite

4. Remote sensing of Earth:

Remote sensing is the science of obtaining information about objects or areas on Earth from space by using satellites. They are used for better understanding of the phenomenon-taking place on the surface of Earth. Images taken from Remote Sensing Satellites facilitate the scientists and

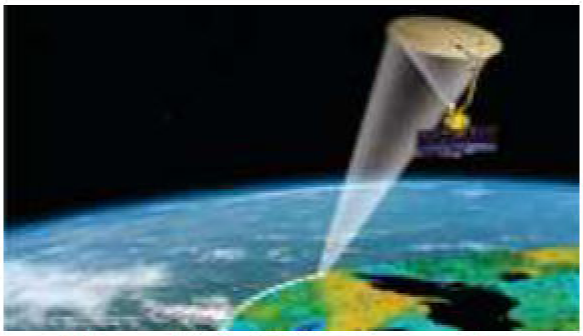


Figure 12.9: A Remote sensing satellite imaging Earth

researchers in studying coastlines, oceans, forests, crops, rivers or natural resources like minerals, oils, gases, etc, hidden under Earth.

✓ Explain that how astronauts survive and research in space.

Outer space is an extreme environment because of no air, less gravity, intense temperature, pressure and hazardous radiations directly coming from the Sun. Such conditions can damage human cells and tissues if exposed for a longer duration. Large space stations have been built in space, which provide basic housing facilities and protection for humans to stay and live in space for a longer period. International Space Station (ISS) is an example of it. For further protection, spacesuits have been designed, which are mandatory for every astronaut to wear when he or she moves out of the space station for work. Spacesuits supply oxygen to the astronauts to breathe while they are in the vacuum of space. They contain water to drink during spacewalks and other Extra Vehicular Activities (EVA). They protect astronauts from being injured by impacts of small bits of space dust. The suits even have visors to protect astronauts' eyes from the bright sunlight. However, inside space station astronauts may not need to wear any spacesuit. Living in a weak gravitational environment may cause human muscles to get weakened. Therefore, astronauts must perform intense workouts on specially designed exercise machines to keep their muscles strong.

Figure 12.10: An astronaut wearing a space suit and standing on the surface of the Moon

✓ Suggest ways to solve the problems that have resulted from space exploration.

Problem of Increasing Space Debris:

Like minerals, water and oxygen on Earth, outer space is also a huge natural reservoir. We need to protect it as we protect the other natural resources on Earth. Sadly speaking, where space exploration is a need of time for the advancement and betterment of our lives, an increase in the launching of spacecrafts, satellites and other space probes, has somehow polluted the nearby space around Earth. This pollution comprises non-operational space junk that remains in space after it has been utilized. It is commonly called Space Debris. Space debris can be as small as a paint fleck or a screw and as large as a fuel tank or even a non-functional satellite, they all are floating in space. Their collision with operational satellites, astronauts or space stations may endanger the safety of future space missions. In this regards, following measures should be taken in order to remove/reduce the space debris.

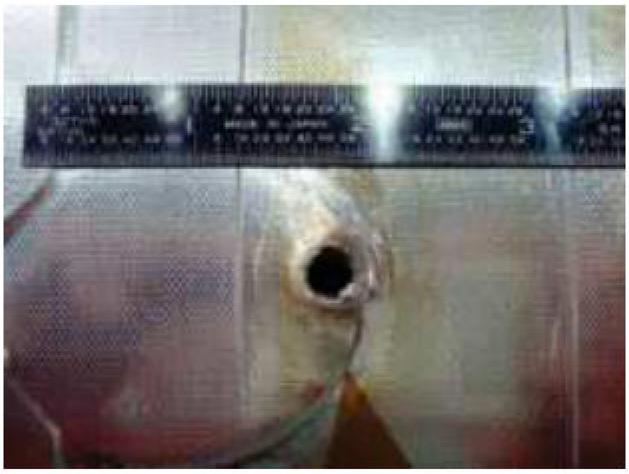


Figure 12.11: Damage caused by space debris on US Space Shuttle

- Minimizing the release of mission-related objects.
- Safeguarding the physical integrity of astronauts, rocket bodies and spacecraft.
- Measures to be taken to reduce the chances of collisions of satellites/space debris.

Long-Term Health Issues:

Upper space is not a habitable place for humans. Even though astronauts wear space suits and live inside a space station to protect themselves against all the dangers, yet a few of the dangers are still inevitable. It has been studied that living in space for

Figure 12.12: Twin brothers Mark and Scott Kelly. Mark spent a year on ISS in space to study long term space travel effects

longer duration may cause genetic changes in human body. This study was carried out on twin brothers; one of whom was kept on Earth, while the other was sent in space to live on International Space Station (ISS) for more than 300 days. The astronaut was brought back to Earth and he was diagnosed with certain changes in his genes i.e. damage to his DNA and reduction in his cognitive abilities.

✓ Identify the technological tools used in space exploration.

Besides space probes, satellites and GPS, following are some of the other technological tools which are used in space exploration.

Satellite Launching Facility (SLF):

Launching a satellite, rocket, a space probe, or even astronauts into space, require a very large facility to be built on ground. This is called a Satellite Launch Facility. It is a technological advancement in itself. Hundreds of scientist and engineers working round the clock made it possible to safely launch space assets.

Figure 12.13: A satellite Launch Facility

Robots:

To overcome the impact of the harsh space environment, scientists have manufactured different types of space robots for their use in space. This includes Fly-Bys, Rovers, Robotic Arms and Orbiters.

Figure 12.14: A robotic arm on ISS

Cameras:

Digital cameras are used with telescopes to take images of objects in space while they are also used with satellites to take high resolution images of the surface of Earth. Navigation and Hazard cameras are used by Satellite Control Stations established on Earth to guide robots, rovers and other digitally operated space probes to study, research and investigate outer space. Microscopic (Camera) is specifically designed for rovers and robots to take pictures of soil and rocks with very high precision to advance the study of planetary geology.

Telecommunication:

Telecommunication is the transmission of images, sound or any other information from one place to another place by using wired or radio systems. The progress in space exploration has made telecommunication devices much more advanced. Today the communication is much faster and reliable as it can transfer a huge amount of data in very short time from a person to any specific location either from ground to ground or from ground to space and backward.

> ✓ Identify new technologies used on earth that have developed as a result of the development of space technology.

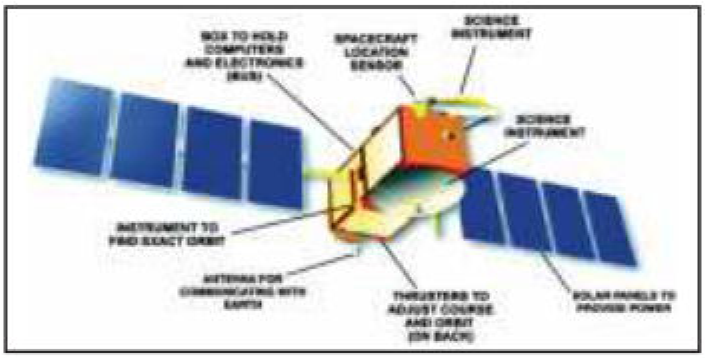
With the passage of time and increasing interest from countries around the world, more and more money is being invested in space exploration which in turn paving our way towards an advanced era of technology. Apart from revolutionizing our medical treatments and health procedures, exploration and research in space has also impacted our daily lives. A few of them are mentioned below:

- **Solar Cells:** They were primarily designed for their use in satellites and space probes. Now, this technology is used as an alternate source of electricity generation in our homes, offices, factories, etc.
- **WiFi**: The concept connecting two remote devices by using a WIFI connection was first applied by the scientists working on a large radio telescope. This modern technology is now being widely used everywhere around the world.
- **Tele-education**: Satellite communication is being used to educate people living in remote and inaccessible areas. This concept is known as Tele-education.
- **Telemedicine**: It allows health care professionals to evaluate, diagnose and treat patients at a distance by using Satellite Communication.

- ✓ Design a space craft and explain its key features of design to show its suitability as a space craft.

Although there are many different parts a satellite or spacecraft consist of but the following are the most fundamental and found in almost every spacecraft.

- **Space Bus:** A box like container to be the body of the spacecraft. It holds the computer and electronics.
- **Solar Panel:** Something to supply electric power.
- **Cameras and other devices:** Some instruments to make scientific measurements or take pictures.
- **Antennas:** Some way to communicate with Earth (both to send data and to receive commands).
- **Micro Thrusters:** Some way to slow down, speed up, or change the direction of the spacecraft to keep it on course or in the right orbit.
- **GPS receiver:** Something to let the spacecraft know where it is and where it is going.



Activity:

By keeping in mind these parts of the satellite/spacecraft, design your own spacecraft. Using cardboard, color papers, scissors and glue make the model of your satellite. Perform following tasks.

- Describe the objectives of your satellite?
- Where do you want to send it? Mars or Moon? Explain your answer.
- Place different parts of satellite/spacecraft on board and explain their working.

SUMMARY

- Telescope is an instrument that helps to see distant objects clearly. There are two types of telescope; refracting and reflecting.
- White Light is the combination of seven different colours.
- A spectroscope splits white light into its seven component colours.
- A spacecraft is a man-made object developed to accomplish a certain task in space.
- Research and development from space exploration has given us several benefits in the field of health, medicines, weather forecasting, navigation, etc.
- Astronauts wear space suits in space.
- SLF, robots, cameras and telecommunication devices are some of the technological tools used in space exploration.
- Advancement due to space exploration is changing our everyday life. Solar cells, Wifi, tele-education and telemedicine are some of the common examples.

EXERCISE

1. Choose the correct answer.

(i) A Spectroscope is used to:
- a) Detect sound waves emitting from a star.
- b) Identify the chemical elements present in a star.
- c) Converge light from a star to a point.
- d) Identify the location of the star.

(ii) A Reflecting Telescope consist of:
- a) A primary and a secondary mirror
- b) Only a single mirror.
- c) Many lenses.
- d) A prism to split light.

(iii) Which Lunar Command Module took humans to the Moon?
- a) Apollo 13.
- b) Apollo 15.
- c) Apollo 16.
- d) Apollo 17.

(iv) MRI or CT scan resulted from the image exploration of:
- a) Saturn.
- b) Moon.
- c) Jupiter.
- d) Sun.

(v) A Global Positioning System (GPS) comprises how many satellites?
- a) More than 15.
- b) More than 20.
- c) More than 30.
- d) Less than 10.

2. Fill in the blanks:

a) The word telescope is a combination of ________ words.

b) Refracting telescopes uses ________.

c) _______ colour has largest wavelength in the spectrum of white light.

d) A spacecraft can be a ___________ or it can be an ___________.

e) MRI is a short form of _____________.

f) Each GPS satellite makes ______ complete orbit in 24 hours.

3. Answer the following questions.

1. Define the following terms:

   a. Telescope.
   b. Refracting telescope.
   c. Reflecting Telescope.
   d. Spectroscope.
   e. Space Exploration.
   f. Space Debris.
   g. GPS.
   h. Remote Sensing.

2. How does a Refracting Telescope differ from a Reflecting Telescope?
3. Explain the construction and working of a Spectroscope.
4. Write down any five benefits of space exploration in the field of health and medicines.
5. What are the different technological tools used in space exploration?
6. Write a short note on the following:

   a. Astronaut surviving in space.
   b. Problems created by space exploration.
   c. Global Positioning System.

7. Write down the names and function of main parts of a satellite/spacecraft.